



Garlock 9800

MATERIAL PROPERTIES*

Color:	Black
Composition:	Carbon fibers with a SBR rubber binder
Fluid Services¹:	Water, saturated steam ³ and inert gases
Temperature², °F (°C)	
Minimum:	-100 (-73)
Continuous Max:	+650 (+343)
Maximum:	+900 (+482)
Pressure², Maximum, psig (bar):	2000 (138)
P x T (max.)², psig x °F (bar x °C)	
1/32 and 1/16":	700,000 (25,000)
1/8":	350,000 (12,000)
Meets Specification:	Fire Safe

PHYSICAL PROPERTIES*

ASTM F36	Compressibility, range, %:	7-17
ASTM F36	Recovery, %:	55
ASTM F38	Creep Relaxation, %:	15
ASTM F152	Tensile, Across Grain, psi (N/mm²):	1500 (10)
ASTM F1315	Density, lbs./ft.³ (grams/cm³):	105 (1.68)
ASTM F433	Thermal Conductivity (K), W/m°K (Btu·in./hr. ft.²·°F):	0.50-0.60 (3.50-4.15)
ASTM F586	Design Factors	1/16" 1/8"
	"m" factor:	3.5 8
	"y" factor, psi (N/mm ²):	2350 (16.2) 3200 (22.1)
ASTM F104	Line Call Out:	F712402A9B2E34K8L302M9 ⁽⁴⁾

SEALING CHARACTERISTICS*

	ASTM F37B Fuel A	ASTM F37B Nitrogen	DIN 3535- 4 Gas Permeability
Gasket Load, psi (N/mm²):	500 (3.5)	3000 (20.7)	4640 (32)
Internal Pressure, psig (bar):	9.8 (0.7)	30 (2)	580 (40)
Leakage	0.1 ml/hr.	0.1 ml/hr.	0.015 cc/min

IMMERSION PROPERTIES* - ASTM F146 Fluid Resistance after Five Hours

	ASTM #1 Oil 300°F (150°C)	ASTM IRM #903 300°F (150°C)	ASTM Fuel A 70-85°F (20-30°C)	ASTM Fuel B 70-85°F (20-30°C)
Thickness Increase, (%)	0-10	15-40	0-10	5-20
Weight Increase, (%)	<20	-	<20	<20
Tensile Loss, (%)	-	<65	-	-

Notes:

This is a general guide and should not be the sole means of selecting or rejecting this material. ASTM test results in accordance with ASTM F-104; properties based on 1/32" (0.8mm) sheet thickness unless otherwise mentioned.

* Values do not constitute specification Limits

¹ See Garlock chemical resistance guide.

² Based on ANSI RF flanges at our preferred torque. When approaching maximum pressure, continuous operating temperature, minimum temperature or 50% of maximum P x T, consult Garlock Applications Engineering. Minimum temperature rating is conservative.

³ Minimum recommended assembly stress = 4,800psi. Preferred assembly stress = 6,000-10,000psi. Gasket thickness of 1/16" strongly preferred. Retorque the bolts/studs prior to pressurizing the assembly. For saturated steam above 150psig or superheated steam, consult Garlock Engineering.

⁴ A9: Leakage in Fuel A (Isooctane), Gasket Load = 500psi (3.5N/mm²), Pressure = 9.8psig (0.7bar): Typical = 0.1ml/hr, Max = 0.5ml/hr. A9: Leakage in Nitrogen, Gasket Load = 3,000psi (20.7N/mm²), Pressure = 30psig (2bar): Typical = 0.1ml/hr, Max = 0.5ml/hr. M9: Tensile Strength = 1,400psi min. (9.7N/mm² min.).